

Requirements Table — Pet Adoption & Shelter Management System

Problem Statement #57 | Media, Events & Community — Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow staff to log animal vaccination and medical intake records, updating animal availability status to 'Ready for Adoption'.	High	Pass: Saving a complete intake record (vaccinations and medical examination) with no open quarantine flag sets the animal's status to 'Ready for Adoption'. Fail: The status becomes 'Ready for Adoption' while the intake record is incomplete or a quarantine flag recorded in it is still open.	Given in problem statement — ensures only medically cleared animals become adoptable.
FR-002	Functional	The system shall allow an adoption applicant to search and filter the pet catalog by species, age, breed, and compatibility with children or other pets.	Medium	Pass: Filtered results display only animals matching all selected criteria. Fail: Results include any animal not matching a selected filter.	Main discovery function; Medium because applicants can still browse the unfiltered catalog.
FR-003	Functional	The system shall allow a logged-in adoption applicant to submit an adopter questionnaire (housing type, household members, other pets, adoption experience — all required) for an animal whose status is 'Ready for Adoption'.	High	Pass: A complete questionnaire creates an application in 'Pending Review', linked to the animal and applicant, that appears in the staff review queue within 5 seconds. Fail: An application is created with a required field missing, for an animal that is not 'Ready for Adoption' or has been removed from the catalog, or as a second open application by the same applicant for the same animal.	Complete, structured adopter data is required before staff can screen an applicant.
FR-004	Functional	The system shall allow shelter staff to record background/reference check results and move an adoption application through its states (Pending Review → Approved or Rejected; Approved → Completed or Rejected), notifying the applicant of each change.	High	Pass: 'Approved' is accepted only when a background-check result is recorded and no other application for the same animal is Approved, and sets the animal to 'Adoption Pending'; 'Rejected' requires a reason code and returns an 'Adoption Pending' animal to 'Ready for Adoption'; 'Completed' sets the animal to 'Adopted' and closes its other open applications as Rejected; the applicant is notified within 1 minute of each change. Fail: Any change happens without its condition, two applications for one animal are Approved, or a notification is not sent within 1 minute.	Formal decision checkpoint; guarantees one adopter per animal and a defined end state ('Adopted').
FR-005	Functional	The system shall allow shelter staff to manage foster home placements: record start date, expected end date and assigned foster caregiver, flag placements past their expected end date, and end a placement.	Medium	Pass: Marking an animal 'In Foster' requires a placement with start date, caregiver and an expected end date later than the start date; a placement past its expected end date appears in the staff overdue list; ending a placement restores the animal's previous status ('In Intake' or 'Ready for Adoption'). Fail: A placement is saved with an end date on or before its start date, an 'In Foster' animal has no active placement, an overdue placement is not flagged, or an ended placement leaves the animal 'In Foster'.	Problem statement requires managing foster placement timelines; Medium because adoptions can run without fostering.

NFR-001	Non-Functional (Performance & Security)	The pet search catalog must support multi-attribute filtering (Species, Age, Breed, Good with Kids / Other Pets) with page response < 200 ms at the 95th percentile, and must expose only public animal fields.	Medium	Pass: A load test with 200 concurrent users running mixed filter queries records a 95th-percentile response < 200 ms, and no catalog response contains medical-intake or applicant data. Fail: 95th-percentile response ≥ 200 ms, or any catalog response includes staff-only fields.	Given in problem statement (200-user peak load is an assumed figure); Medium because a slower catalog degrades but does not block adoption.
NFR-002	Non-Functional (Security & Privacy)	The system shall require login for all functions except the public catalog, restrict medical-intake records and background-check results to Shelter Staff, and restrict each adopter questionnaire to Shelter Staff and the applicant who submitted it.	High	Pass: An unauthenticated request to any protected record returns HTTP 401; an applicant requesting medical, background-check or another applicant's questionnaire data receives HTTP 403; an applicant can view their own questionnaire. Fail: Any user reads or edits data outside these rules.	Protects sensitive animal medical data and applicants' personal information.

Actors: Adoption Applicant, Shelter Staff.

Priority basis: High = needed for a safe, working adoption cycle (intake, application, decision, data protection); Medium = improves usability or operations, but the adoption cycle still works without it.